

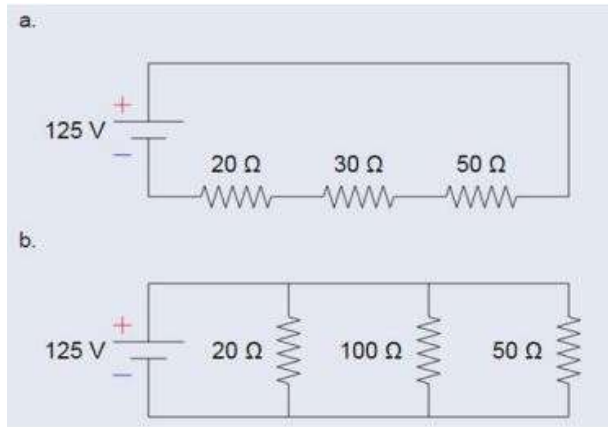
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Lab Exercise: Schematics with ESP32

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Part A



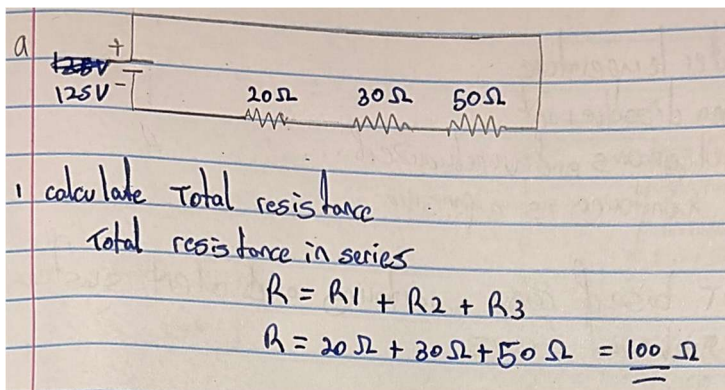
For each of these circuits:

1. Calculate the total resistance in ohms,

Circuit A

Total Resistance In Series = $R_1 + R_2 + R_3$

$$R = 20\Omega + 30\Omega + 50\Omega = 100\Omega$$

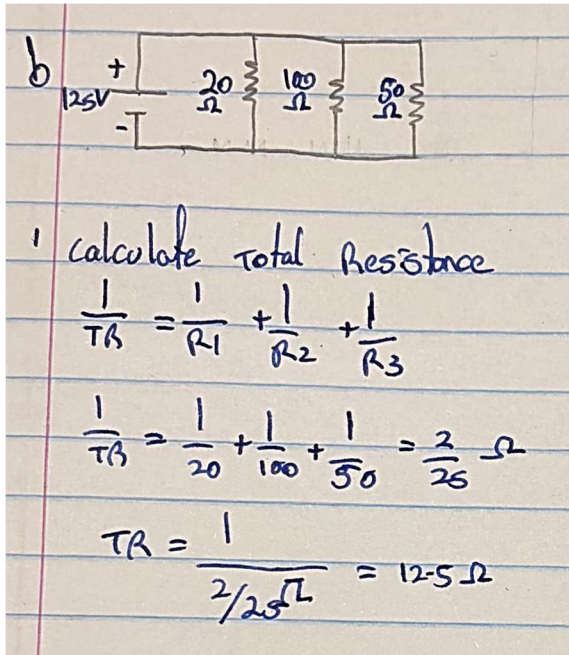


Circuit B

Total Resistance In Parallel = $1/R_1 + 1/R_2 + 1/R_3$

$$\frac{1}{R} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{2}{25}\Omega = 0.08\Omega$$

$$R = \frac{1}{0.08} = 12.5\Omega$$

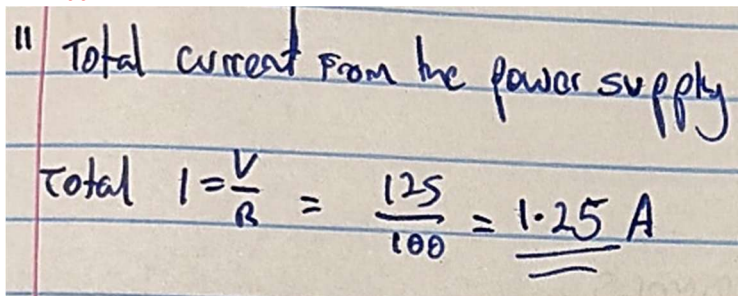


2. Calculate the total current from the power supply,

$$I = \frac{V}{R}$$

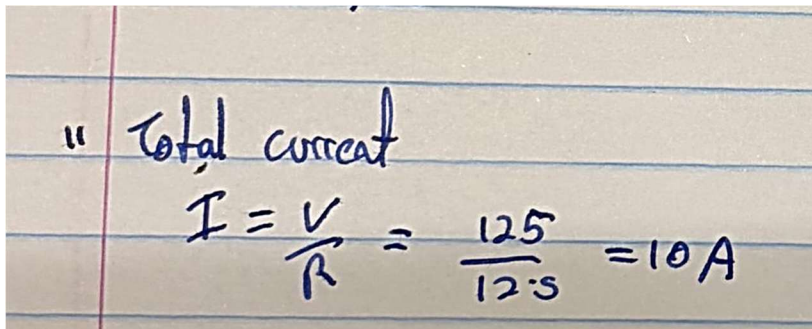
Circuit A

$$I = \frac{125V}{100\Omega} = 1.25A$$



Circuit B

$$I = \frac{125V}{12.5\Omega} = 10A$$



3. Calculate the current through each resistor

Circuit A

In series circuit A the current will be same at all points of the circuit so meaning all resistors will have 1.25 A passing through them

iii calculate current through each resistor.
in series circuits current is same every point so 1.25 at all resistors.
 $I_1 = I_2 = I_3 = \underline{1.25 A}$

Circuit B:

$$20\Omega: I = \frac{125}{20} = 6.25 A$$

$$100\Omega: I = \frac{125}{100} = 1.25 A$$

$$50\Omega: I = \frac{125}{50} = 2.5 A$$

$$\text{Total} = 6.25 + 1.25 + 2.5 = 10 A$$

iii current through each resistor
 $20\Omega \Rightarrow I = \frac{125}{20} = \underline{6.25 A}$
 $100\Omega \Rightarrow I = \frac{125}{100} = \underline{1.25 A}$
 $50\Omega \Rightarrow I = \frac{125}{50} = \underline{2.5 A}$
 $6.25 + 1.25 + 2.5 = \underline{10 A}$

In Parallel Circuit is not the same at all points of the circuit but adds up to the total current which is 10 A

- Calculate the Voltage drop through each resistor

Circuit A

In series circuit the voltage is different at all points of the circuit

iv Voltage drop across each resistor.
 $V = IR$
 $20\Omega \rightarrow V = 1.25 \times 20 = 25V$
 $30\Omega \rightarrow V = 1.25 \times 30 = 37.5V$
 $50\Omega \rightarrow V = 1.25 \times 50 = 62.5V$
 $25 + 37.5 + 62.5 = \underline{125V}$

$$20\Omega: I = 1.25 \times 20 = 25V$$

$$30\Omega: I = 1.25 \times 30 = 37.5V$$

$$50\Omega: I = 1.25 \times 50 = 62.5V$$

$$\text{Total} = 25 + 37.5 + 62.5 = 125 V$$

Circuit B

The voltage across parallel circuits is the same at all points so across each resistor the voltage will be 125 V

V Voltage across circuit B
voltage remains the same at all points of parallel circuits so $V_1 = V_2 = V_3 = 125V$

5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)

Circuit A:

$$20\Omega: P = 25 \times 1.25 = 31.25W$$

$$30\Omega: P = 37.5 \times 1.25 = 46.875W$$

$$50\Omega: P = 62.5 \times 1.25 = 78.125W$$

V Power through each resistor.
20 Ω resistor $\rightarrow P = VI = 25 \times 1.25 = 31.25W$
30 Ω resistor $\rightarrow P = 37.5 \times 1.25 = 46.875W$
50 Ω resistor $\rightarrow P = 62.5 \times 1.25 = 78.125W$

Circuit B

$$20\Omega: P = 125 \times 6.25 = 781.25W$$

$$100\Omega: P = 125 \times 1.25 = 156.25W$$

$$50\Omega: P = 125 \times 2.5 = 312.5W$$

B Power across all resistors
20 $\Omega \rightarrow 125 \times 6.25 = 781W$
100 $\Omega \rightarrow 125 \times 1.25 = 156.25W$
~~50~~ 50 $\Omega \rightarrow 125 \times 2.5 = 312.5W$